An Innovative Pipe Inspection Tool Utilizing Electromagnetic Resonance Coupling and Machine Learning

Tarek M. Mostafa, Guang Ooi, Mehmet Burak Ozakin, *Member, IEEE*, Moutazbellah Khater, Mohamed Larbi Zeghlache, Hakan Bagci, *Senior Member, IEEE*, and Shehab Ahmed, *Senior Member, IEEE*

***Abstract*—This article describes an advanced tool that uses electromagnetic resonance coupling and machine learning techniques to detect and characterize metal loss on the inner surface of a metallic pipe. The proposed tool uses a transmitter coil placed along the axis of the pipe and four sensor coils installed around the transmitter coil. Any defect on the pipe surface leads to changes in the impedance of the transmitter and sensor coils as well as in the mutual coupling between them, thus creating a detectable variation in the outputs of one or multiple sensor coils. An artificial neural network is developed to reconstruct two-dimensional pipe cross-sections and to completely characterize the defects using these variations. The proposed tool is tested and validated via simulations, and data are collected using an experimental prototype. Results show that the tool can fully characterize the size, location (azimuthal angle), and level (thickness) of metal loss.**

***Index Terms*—Eddy current, long short-term memory (LSTM), machine learning (ML), neural networks (NNs), pipeline inspection, resonance coupling, wellbore integrity.**

I. Introduction

STEEL pipes are frequently utilized in various applications, such as wellbore tubing, which is essential for extracting oil and gas, and pipelines, which are essential for transporting them. Typically, pipelines are buried underground and surrounded by external forces from soil weight and other loads [1]. These forces may cause the formation of cracks in pipes [2]. Moreover, environmental factors, such as humidity and temperature, may cause corrosion and deformation, which may lead to metal loss and cracks. Furthermore, oil wells are frequently subjected to considerable levels of stress and highly corrosive fluids. These factors exacerbate corrosion and deformation of wellbore tubing, thus resulting in either partial metal loss or cracks on the external and/or internal walls of the tubing [3]. Cracks in pipelines and wellbore tubing cause severe economic loss, environmental pollution, and fatal accidents [4]. Thus, it is essential to monitor pipelines and well tubing continuously to ensure that all production and transportation processes are safe, efficient, and environmentally friendly.

Multiple nondestructive testing (NDT) methods [5] have been developed to inspect well tubing and pipelines. These methods can be categorized into three main groups [6], [7] ultrasonic tools [8], mechanical tools (e.g., calipers) [9], and electromagnetic (EM) tools [5], [10].

Ultrasonic testing (UT) is an effective NDT method; yet, it has several disadvantages. It requires a coupling medium, often a liquid, to promote energy transfer between the sensors and the pipe wall. Furthermore, the pipe wall should be clean because ultrasonic scattering from rough surfaces causes interference with the primary signal used for inspection.

Mechanical calipers sweep the pipe's inner wall with measurement fingers (or feelers). The major limitation of this method is the presence of blind areas between fingers where defects may be overlooked [6].

EM NDT methods are often more cost-effective. Remote field eddy current (RFEC) testing [11], pulsed eddy current (PEC) testing [12], metal magnetic memory testing [13], and magnetic flux leakage (MFL) testing [14] are examples of these methods. MFL is the most commonly used EM-based technique. It is a fast inspection method but requires strong magnet(s); hence, MFL-based inspection tools are usually heavy, bulky, and power hungry. Furthermore, MFL fails to distinguish between inner and outer losses. Generally, the complexity, weight, and size of the existing EM inspection tools are major limitations for multiple applications. To address these challenges, this work proposes a framework that relies on magnetic resonant inductive coupling (MRIC) and machine learning (ML) methods. This framework is employed to develop an innovative, simple, and relatively lightweight solution that overcomes the aforementioned

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Tarek M. Mostafa, Guang Ooi, Mehmet Burak Ozakin, Moutazbellah Khater, Hakan Bagci, and Shehab Ahmed are with the King Abdullah University of Science and Technology, Thuwal 23955, Saudi Arabia (e-mail: tarek.mostafa@kaust.edu.sa; guang.ooi@kaust.edu.sa; mehmet.ozakin@kaust.edu.sa; moutazbellah.khater@kaust.edu.sa; hakan.bagci@kaust.edu.sa; shehab.ahmed@kaust.edu.sa).

Mohamed Larbi Zeghlache is with the Saudi Aramco Advanced Research Centre, Dhahran 31311, Saudi Arabia (e-mail: mohamed.larbizeghlache.1@aramco.com).

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TABLE I
COMPARISON OF THE MOST USED INSPECTION TECHNIQUES AND THE PROPOSED TECHNIQUE

Features \ Tech.	Mechanical Caliber	MFL	UT	RFEC	PEC	Proposed Technique
Physical Size	Medium	Large	Small	Large	small	small
Power Requirements	Low	High	Medium	High	Low	Very Low
Complexity	Simple	Complex	Complex	Complex	Complex	Simple
Inner loss detection	Limited	Possible	Possible	Possible	Possible	Possible
Outer loss detection	Limited	Possible	Possible	Possible	Limited	Limited
Full loss detection	Limited	Possible	Possible	Possible	Possible	Possible
Array of sensors required (Local detection only)	NA	Required	Required	Not required	Required	Not required
Coupling medium required	NA	Not required	Required	Not required	Not required	Not required
Contact with pipe wall	Required	Required	Not required	Not required	Not required	Not required
Defect quantification capability	Limited	Medium	High	Limited	High	High
Pipe material limitation	NA	Ferromagnetic pipes only	Metal pipes	Mainly ferromagnetic	Mainly non-ferromagnetic	Metal pipes

limitations of the existing EM tools while providing comprehensive loss characterization. Table I provides a comparison of different inspection techniques.

After its rediscovery in 2007, MRIC has been widely used in wireless power transfer (WPT) applications [15]. MRIC systems are designed to effectively exchange energy by forming a strong magnetic coupling between two or more coils that are set to resonate at a specific frequency. Furthermore, magnetic fields from the surroundings, dc fields, and magnets do not affect MRIC systems. Its implementation as a part of the framework proposed in this study comprises one transmitter coil and four sensor coils tuned to the same resonating frequency.

ML methods, including neural networks (NNs), have made it possible to effectively manage the complexity of Big Data sets by "learning" their intrinsic properties [16], [17], [18]. In this work, a computational framework is developed to incorporate an artificial neural network (ANN) and a state-of-the-art finite-difference time-domain (FDTD) method. The ANN acts as the inverse solver, i.e., it reconstructs cross sections of the pipe under inspection and completely characterizes metal loss from EM field measurements. The forward solver FDTD computes EM fields (at sensor coil locations) for a given defect or metal loss scenario. This forward solver can efficiently simulate numerous scenarios in a short time period and is therefore used to train the ANN.

The benefits of the proposed approach are demonstrated in the following three different ways.

1) It is used to predict the cross section of a pipe (with metal loss) from synthetically generated sensor readings (using the commercial EM solver ANSYS).
2) A prototype is built, and various pipe cross-sections are reconstructed from the measured sensor readings.
3) A workflow is built to utilize long short-term memory recurrent neural networks (LSTM-RNNs) to extract features from small amounts of spatially oriented sensor data, visualizing the pipe structure with pixel-level accuracy, and mapping each pixel's value from input sensor data to represent the probability of defects.

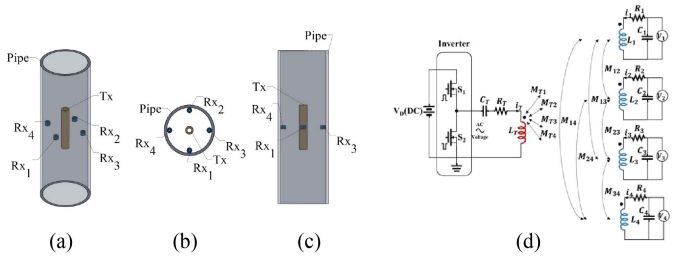


Fig. 1. Pipe inspection tool. (a) Trimetric view. (b) Top view. (c) Front view. (d) Equivalent circuit model.

II. SYSTEM STRUCTURE AND OPERATING PRINCIPLE

A simplified version of the proposed tool is illustrated in Fig. 1(a)–(c), where one transmitter coil (Tx) and four sensor coils (Rx_1–Rx_4) are identified. The transmitter coil is axially aligned with the pipe. In contrast, sensor coils are circumferentially distributed with 90° spacing at equal radial distances from the center of the transmitter coil. To achieve the best detectability possible, the sensor coils are located as close as possible to the inner surface of the pipe. Both transmitter and sensor coils comprise air-core multilayer solenoids. Air-cored coils were selected rather than magnetic-cored coils for their low cost, linear dependence on frequency and temperature, and ease of adjustment. Adjusting the size of coils for optimal sensitivity to defects is possible; however, this falls beyond the purview of this research and will be discussed in a separate context.

Fig. 1(d) shows an equivalent circuit model of the proposed inspection system. The system is divided into the transmitting and sensing sides. On the transmitting side, switches S_1 and S_2 form a half-bridge inverter to convert the dc input voltage (V_D) into an ac signal. This ac signal is then fed to the transmitting coil (L_T) through the series resonant capacitor (C_T) and transmitter side resistance (R_T). R_T is the equivalent series resistance (ESR) of L_T. The series compensation topology is used on the transmitter side as it requires a voltage source.

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Therefore, the half-bridge inverter can be directly used without any extra components that will unnecessarily complicate the design. Furthermore, the sensing side comprises four parallel resonant circuits. Each resonant circuit consists of a resonant capacitor C_i ($i = 1, 2, 3, 4$) in parallel with a sensor coil (L_i) and an ESR (R_i). Sensor coils are coupled to the transmitter coil through magnetic mutual coupling coefficients (M_{Ti}); for example, L_T and L_1 are coupled via M_{T1}. The four sensor coils are assumed to be identical and coupled with coupling coefficients M_{ij} ($j = 1, 2, 3, 4$), e.g., L_1 and L_2 are coupled via M_{12}. The output voltages from the sensing resonant circuits are then measured, recorded, processed, and used as inputs to the ANN.

As mentioned earlier, capacitances are selected to provide the same resonant frequency. Consequently, the straightforward means in (1) are considered to produce resonant coupling on both sides

$$\omega = \frac{1}{\sqrt{L_T C_T}} = \frac{1}{\sqrt{L_i C_i}} \tag{1}$$

where ω is the operating angular switching frequency that is equal to $2\pi f$.

The circuit model shown in Fig. 1(d) is analyzed according to Kirchhoff's voltage law (for details, refer to Appendix A), assuming the sinusoidal approximation [19], whereas harmonics of the switching frequency are neglected and the tank waveforms are assumed to be purely sinusoidal. Accordingly, the output voltage from the sensing coils is obtained from

$$V_i = \frac{-I_i}{j\omega C_i} = \frac{j\omega\left(M_{Ti} I_T + M_{i1} I_1 + M_{i2} I_2 + M_{i3} I_3 + M_{i4} I_4\right)}{j\omega C_i R_i - \omega^2 C_i L_i + 1} \tag{2}$$

where $M_{ij}\,|_{(i=j)} = 0$.

As per Faraday's law of EM induction states, the magnetic field generated by a transmitter coil induces an eddy current in the walls of the conductive pipe. This current, in turn, produces an induction EM field. This secondary field and the direct field from the transmitter induce a voltage in the sensor coils. The presence of a defect alters the eddy current density distribution on the pipe wall, thus influencing the EM field and consequently the voltage induced in the sensor coils. This can be interpreted as changes in the inductance and mutual inductance values. The pipe properties, in addition to the number, type, size, and position of the defects, directly affect the value of each sensor coil as well as the mutual coupling between the system coils. Therefore, the resonance condition for each circuit and the output voltage from each sensor differ. The diagnosis and characterization of defects are made possible by monitoring and measuring these changes. Analytically calculating the above-mentioned inductance and mutual inductance changes is not a trivial task. Hence, a numerical solution can be utilized. In this study, a computationally efficient FDTD EM solver is used to train the ANN. ANSYS Maxwell solver and experimental results are used to test and validate the proposed approach (see Fig. 2).

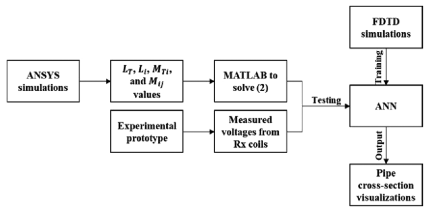
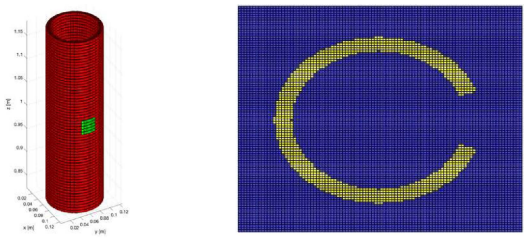


Fig. 2. Flowchart of the implemented framework.



(a) (b)

Fig. 3. (a) Discretized problem space (red and green cells represent the metallic pipe and rectangular hole regions, respectively). (b) Two-dimensional cross section of the problem space with the inspection tool (yellow and blue cells represent the metallic pipe and background, respectively).

III. Forward Solver

This section describes the EM-based forward solver that is used to generate synthetic data to train the ML algorithm for metal loss prediction. In this study, the FDTD method is employed because of its simple implementation [20]. However, because the excitation frequency is extremely low, an extended version of the traditional FDTD scheme is used to accommodate large time step sizes without sacrificing stability [21]. This is the so-called quasi-static FDTD (QS-FDTD) method that scales up the dielectric permittivity (ε) by a coefficient α, thereby increasing the numerical time step to $\Delta\, t^{\mathrm{QS}} = \sqrt{\alpha}\,\Delta t$. This approach reduces the total computational time. Hence, in this study, the Maxwell's equation-based three-dimensional QS-FDTD method is used to generate training data for ML. Readers are referred to [22] for additional details about the formulation and implementation of the QS-FDTD method.

A. Numerical Model of the Pipe Inspection Tool

The numerical model used in training the ANN is the discretized version of the EM inspection tool shown in Fig. 1. On the circular transmitter coil, only the x and y current components are considered by QS-FDTD. Multiturn and multilayer coils can be similarly discretized. This current model for the coils generates a magnetic field with a dominant component in the z-direction.

An example simulation configuration is shown in Fig. 3(a). The problems involve the aforementioned inspection tool, which is modeled as a metallic cylindrical pipe with holes. The pipe is 2.0 m long, and its inner and outer radii are 51.13 and 57.15 mm, respectively. Fig. 3(b) shows a 2-D cross section of the computation domain. Rather than modeling the sensor coils,

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TABLE II
METAL LOSS RANDOM INITIALIZATION PARAMETERS

Parameter	Metal loss count	Width (mm)	Length (mm)	Thickness (mm)	Location (°)	Total cases
Value	1–4	1–60	1–60	1–6	0–360	80 000

the z-component of the magnetic field at points where these coils would be centered is computed.

The dimensions of the grid cell used in the QS-FDTD are $\Delta x = \Delta y = 1$ mm and $\Delta z = 6$ mm. Thus, there are a total of $125 \times 125 \times 334 = 5\,218\,750$ cells in the whole computation domain. The first-order absorbing boundary condition [20] was used on the outer boundaries of the computation domain (to approximate the open-space problem). The operating frequency of the transmitter coil is set at 100 kHz. The conductivities of the background/metal loss and metal pipe are 0.1 S/m and 5.7×10^6 S/m, respectively. The amplitudes of the voltages induced on the receivers are determined [23] from the maximum value of the frequency-domain response of the time-domain signal (QS-FDTD produces time-domain fields). All QS-FDTD simulations are executed in MATLAB, and the total computation time for a typical simulation is ∼30 min on an i9 CPU running at 2.4 GHz.

IV. MACHINE LEARNING

This section describes the ML workflow used to predict metal loss scenarios from the data collected at the sensors.

A. Simulation Scenarios

In this work, defects are configured as complete losses (holes), partial losses on the inner surface of the wall, or combinations of the two. For all simulations, a random number of individual metal losses are developed on the pipe, with the first metal loss having a 100% chance of appearing; the second, a 75% chance; the third, a 50% chance; and so on. Note that multiple defects are allowed to overlap to create additional complex defect shapes with varying levels (thickness) of metal loss.

The simulations are run in parallel on multiple workers of a cluster of nodes on supercomputer Shaheen II [24]. A batch of 700 simulations is executed on Skylake, Cascade Lake, and AMD Rome CPUs with an average completion time of ∼6 to 8 h. A total of 80 000 different scenarios are simulated by varying parameters, such as loss number, size, and thickness within specific ranges. The results are compiled into a dataset (see Table II). Subsequently, 60%, 20%, and 20% of this dataset are randomly selected to train, validate, and test the ANN, respectively. The validation dataset is used as an indicator for overfitting during the training process and is randomly interchanged with the training data during each training epoch. The testing data are unseen by the ANN in the training process and provide indications of the generalizability and accuracy of the ANN. The whole dataset is then used to train the ANNs once the testing is complete, and an optimal ANN structure is identified.

B. Data Representation and ANNs

At each layer, the sensors are evenly placed at equal distances on a circular holder. Voltages induced in the coils are recorded as $S^m = [s_1, s_2, \ldots, s_N]^m$, where m is the measurement taken at the mth layer and s_n is the nth sensor voltage ($n = 1, 2, \ldots, N$) on the circular holder. At the start of inspection tests, an array of reference values R is experimentally acquired for a known section of the pipe where there is no metal loss. The differences between the reference values R and S^m are recorded in a difference array D^m

$$D^m = R - S^m. \tag{3}$$

D^m is used rather than S^m to cancel the linear shifting effects on sensor readings due to a variety of external factors, such as the varying EM properties of pipes, the history of their use, and hardware specifications. D^m taken at layers $m = 12, \ldots, M$ are stacked to create difference matrix D for the inspected length of the pipe. This matrix is used as an input to the ANN. Each row of D is used by the ANN to generate a cross-sectional image of the corresponding measured layer m. The depth of the defects resulted in various D^ms in all sensors. This combination of D^ms then constitutes unique defect signatures.

A feasible approach for representing the pipe cross-sections is to label all empty space, including metal loss elements, as zeros, while labeling the steel elements as ones. The mapping of the complete cross section is then fed into the ANN, which predicts a value between zero and one for each pixel on the 2-D map. Each element on the map possesses a unique set of coefficients learned by the ANN as input-to-pixel transformations. The value of each pixel may be considered as the certainty of the prediction as to whether it represents steel or loss (empty space).

C. ANN Structure

To address the long-term dependencies in the sensor voltages, an LSTM-RNN, which comprises N individual cells in parallel, was deployed [25]. The memorial nature of such cells is represented by the input memory c^{t-1} and output memory c^t. Each cell in the network could throttle the amount of memory to be retained using a single-layer NN into which the memory state c^{t-1} and output h^{t-1} from the previous cell, in addition to the current input x^t, are fed. The LSTM-RNN cell has three gates [26] the forget gate, the update gate, and the output gate, as shown in Fig. 4.

LSTM-RNNs employ specialized gate units to regulate the gradients of loss functions to prevent vanishing and exploding gradients. The forget gate can be formulated as follows:

$$\Gamma_f^t = \sigma\left(W_f\left[h^{\langle t-1\rangle}, x^{\langle t\rangle}\right] + b_f\right) \tag{4}$$

where σ is a sigmoid activation function; W_f and b_f are the weight matrix and bias vector of the forget gate, respectively; and $[h^{\langle t-1\rangle}, x^{\langle t\rangle}]$ represents the concatenation of the output from the previous cell and current input. This concatenated matrix undergoes elementwise multiplication with W_f. Similarly, the update and output gates, respectively, can be formulated as

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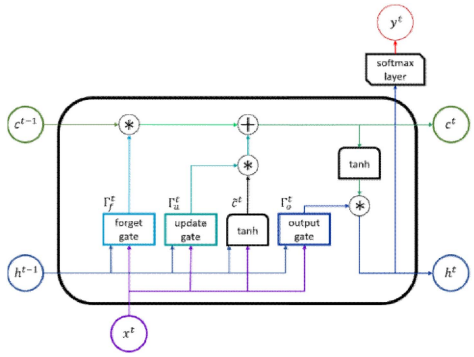


Fig. 4. Components of a single LSTM-RNN cell unit.

TABLE III
RANGE OF ANN HYPERPARAMETER VARIATIONS

Hyperparameter	Variations	Range of n
Number of neurons per layer	2^{n+2}	1:8
Number of dense layers	n	1:8
Dropout ratio	$0.1n$	1:8
Batch size	2^{n+2}	1:8
Learning rate	$0.0005n$	1:8
Optimizer	Adam, Adagrad, SGD, RMSprop	

follows:

$$\Gamma_o^t = \sigma\left(W_o\left[h^{\langle t-1\rangle}, x^{\langle t\rangle}\right] + b_o\right) \tag{5}$$

and

$$\Gamma_u^t = \sigma\left(W_u\left[h^{\langle t-1\rangle}, x^{\langle t\rangle}\right] + b_u\right). \tag{6}$$

Additionally, a candidate value $\tilde{c}^t$ is computed as follows:

$$\tilde{c}^t = \tanh\left(W_c\left[h^{\langle t-1\rangle}, x^{\langle t\rangle}\right] + b_c\right). \tag{7}$$

Finally, the memory and output of the cell are calculated as follows:

$$c^t = \Gamma_u^t * \tilde{c}^t + \Gamma_f^t * c^{t-1} \tag{8}$$

$$h^t = \Gamma_o^t * c^t. \tag{9}$$

The last cell of the LSTM-RNN is connected to a series of dense layers. The final output layer is an $N^2 \times 1$ array of neurons, whose elements represent the pixels in a cross-sectional profile of size $N \times N$. A grid search approach has been adopted to optimize the hyperparameters [27] of the ANN in this study. The investigated ANN hyperparameters, as well as their range of values, are listed in Table III. The performance of the ANNs resulting from each set of hyperparameter values is evaluated by calculating the average mean absolute errors (MAEs) of the ANNs on the test dataset (the MAEs are shown in Appendix B). The optimal set of hyperparameters, which allows the ANN to produce the lowest MAE on the test dataset, is shown in Table IV. The structure of the ANN, constructed with the optimal hyperparameters, is shown in Table V.

The values of each pixel represent its likelihood of being a pipe body or an empty space element. When connected in series, the information from any individual sensor can be memorized

TABLE IV
HYPERPARAMETERS FOR OPTIMAL ANN PERFORMANCE

Hyperparameter	Value
Number of neurons per layer	256
Number of dense layers	3
Dropout ratio	0.7
Batch size	64
Learning rate	0.002
Optimizer	Adam

TABLE V
OPTIMIZED ANN STRUCTURE

Layer	Output Shape	Number of Parameters
Input	4	N/A
LSTM	256	264 192
Dense	256	65 792
Dense	256	65 792
Dense	256	65 792
Dropout	256	0
Output	10 000	2 570 000
Total Parameters	3 031 568	

by simply setting their forget gate coefficient to one. For every pixel in the prediction task, the LSTM-RNN corrects its weight matrices and biases such that the best fit for the training dataset can be reported. Such values are updated in the backpropagation step of the training process and optimized using the adaptive moment (ADAM) algorithm.

In this study, to compensate for measurement noise, zero-mean additive white Gaussian noise (AWGN) at multiple levels is injected into the training dataset. Five ANNs—one with the original output values, two trained with datasets with AWGN with 1 sigma, and two more trained with AWGN with 2 sigma—are trained with the resulting dataset. During testing, noise was introduced to the test dataset; the input was fed into the ANNs in parallel. An arbitrary threshold is set for the confidence score for the cross sections predicted by the ANNs, where all eligible predictions are averaged and used as the final result.

V. SIMULATION AND EXPERIMENTAL VERIFICATION

The system is modeled, and data are obtained from an EM field solver, particularly ANSYS Maxwell, to demonstrate the practicality of the proposed inspection tool and evaluate the reliability of the ANN to forecast and describe different types of defects. The values of the transmitter and sensor coil inductances, in addition to the mutual inductances between them, are input into MATLAB to solve (2) for each sensor coil. The ANN receives the output (voltages) from the four sensor coils. A "proof of concept" prototype is also developed in addition to the simulated model to test the proposed system on real pipes with flaws.

A. Simulation Verification

The system depicted in Fig. 1 is modeled in ANSYS Maxwell as shown in Fig. 5. The model features are summarized in Table VI. For ease of understanding the simulated cases, the location of sensor coil Rx_1 is referenced at 0°, whereas the

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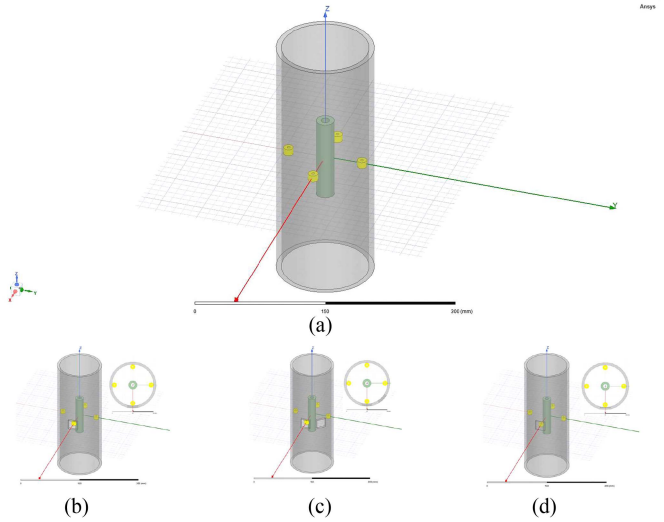


Fig. 5. Three-dimensional view of the modeled system in ANSYS Maxwell for different cases. (a) No loss. (b) Single full metal loss. (c) Double full metal loss. (d) Inner metal loss.

TABLE VI
MODEL SPECIFICATIONS

Parameter	Value
Pipe conductivity (MS/m)	5
Pipe relative permeability	100
Pipe inner radius (mm)	51.13
Pipe thickness (mm)	6.02
Transmitter coil length (mm)	100
Transmitter coil number of Turns	409
Transmitter coil inner diameter (mm)	10
Transmitter coil outer diameter (mm)	21.2
Sensor coil length (mm)	8
Sensor coil number of turns	470
Sensor coil inner diameter (mm)	3.43
Sensor coil outer diameter (mm)	11.43
Frequency (kHz)	100
DC input (V)	32

locations of sensor coils Rx_2, Rx_3, and Rx_4 are at 180°, 90°, and 270°, respectively. Frequently, WPT (Qi standard) uses a 100-kHz operating frequency for achieving optimal performance and ultimate power transmission. Moreover, it allows for the use of smaller coils and resonance capacitors, thus resulting in an efficient and compact system. Because of its widespread adoption and proven performance in various applications, it is also used in this study.

Table VII lists the details of 18 simulated cases. In the first five cases, the defect size is changed while the defect location is maintained in front of sensor coil Rx_1. In the next five cases, the location of a single defect of size 25 × 25 mm is varied. For cases 11–14, two defects have the same circumferential position. In the last four cases, partial metal loss with different depths and location angles is simulated.

In each case, the sensor readings calculated by MATLAB are injected with zero-mean Gaussian noise of variance equal to 100% of the minimum change among the four sensors. Noise was then added to assess the robustness of the NN. The variance

TABLE VII
SIMULATED CASES

Case Num.	Num. of Defects	Defect Type	Defect Dimensions (mm)	Defect Depth (mm)	Defect Location
1	1	Full Loss	12.5 × 12.5	6.02	0°
2	1	Full Loss	18.75 × 18.75	6.02	0°
3	1	Full Loss	25 × 25	6.02	0°
4	1	Full Loss	31.25 × 31.25	6.02	0°
5	1	Full Loss	37.5 × 37.5	6.02	0°
6	1	Full Loss	25 × 25	6.02	15°
7	1	Full Loss	25 × 25	6.02	30°
8	1	Full Loss	25 × 25	6.02	135°
9	1	Full Loss	25 × 25	6.02	240°
10	1	Full Loss	25 × 25	6.02	345°
11	2	Full Loss	25 × 25	6.02	0° and 135°
12	2	Full Loss	25 × 25	6.02	0° and 180°
13	2	Full Loss	25 × 25	6.02	0° and 240°
14	2	Full Loss	25 × 25	6.02	0° and 270°
15	1	Partial Inner Loss	25 × 25	1	0°
16	1	Partial Inner Loss	25 × 25	2	45°
17	1	Partial Inner Loss	25 × 25	3	0°
18	1	Partial Inner Loss	25 × 25	4	45°

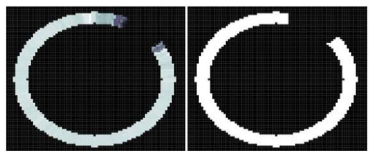


Fig. 6. Images of the cross section predicted by the LSTM-RNN using receiver values from a 25-mm-wide complete metal loss at 135°. (a) Original values. (b) Filtered values.

of the Gaussian noise is set as the smallest sensor change since experimental observations have demonstrated that this value provides generalizable and robust NNs, whereas setting the variance to the largest sensor change has resulted in failed training processes.

The outputs of the NNs are arranged in images of 100 × 100 pixels, as shown in Fig. 6(a). The white and pale blue segments are the pipe body elements predicted with high confidence (with values close to one), whereas the faded pixels, especially in proximity to the suspected metal loss locations, are indications of low certainty. A threshold is applied where all pixels above an arbitrary value are considered metal, whereas the rest are classified as empty space (metal loss), as shown in Fig. 6(b).

Since the generated cross-sectional images are required to accurately represent both the size and location of metal losses, three types of metrics are used to analyze the quality of the predicted images: angle error δ_θ, size error δ_σ, and MAE. The azimuthal angle of metal loss θ on the cross-sectional image is calculated as follows:

$$\theta = \tan^{-1}\frac{y}{x} \tag{10}$$

where x and y are the coordinates of the center of the metal loss. The x- and y-axes in this case are the horizontal and vertical

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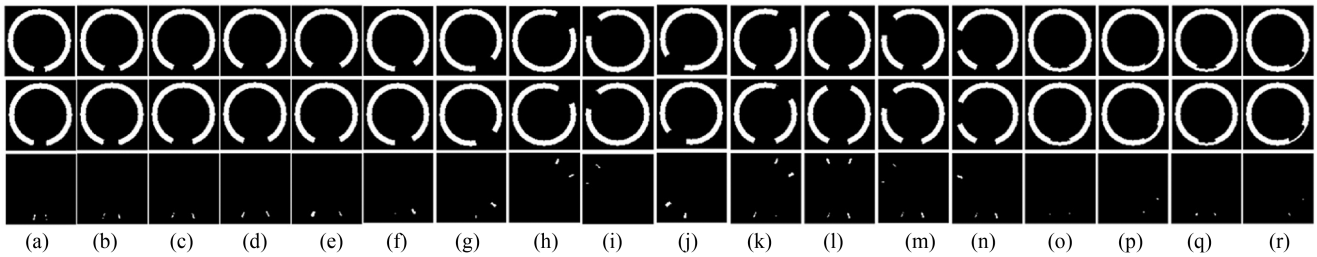


Fig. 7. Performance of the dense NN on hole(s) size, location, and depth estimation for simulated (a) case 1, (b) case 2, (c) case 3, (d) case 4, (e) case 5, (f) case 6, (g) case 7, (h) case 8, (i) case 9, (j) case 10, (k) case 11, (l) case 12, (m) case 13, (n) case 14, (o) case 15, (p) case 16, (q) case 17, and (r) case 18. The first row shows the ground truth; the second row shows the NN prediction; and the last row represents the error pixels.

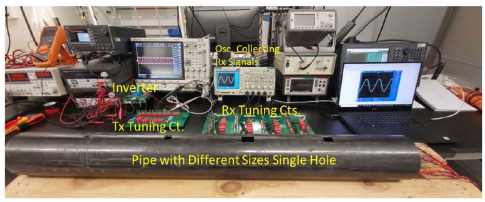


Fig. 8. Experimental setup to verify the introduced inspection tool.

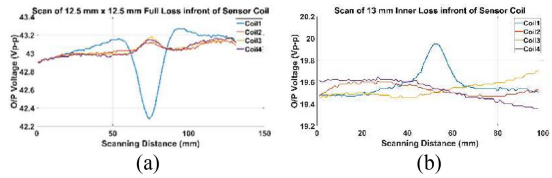


Fig. 9. Examples of readings of sensor coils when (a) full loss and (b) inner loss are detected.

axes of the predicted cross section, respectively. The size of a predicted metal loss is calculated as follows:

$$S = \sum p_h \tag{11}$$

where p_h is the values of pixels that are predicted as metal loss elements. Hence, the angle error and size error can be calculated as follows:

$$\delta_\theta = \frac{\tilde{\theta} - \theta}{n} \tag{12}$$

$$\delta_S = \frac{\tilde{S} - S}{S} * 100\% \tag{13}$$

where $\tilde{\theta}$ is the sum of the predicted metal loss angles, θ is the original metal loss angles, n is the number of holes, $\tilde{S}$ is the sum of the predicted metal loss sizes, and S is the sum of the original metal loss sizes. Finally, the MAE can be calculated as follows:

$$\text{MAE} = \frac{1}{n} \sum_{1}^{n} |\tilde{y}_i - y_i| \tag{14}$$

where $\boldsymbol{n}$ is the total number of pipe body pixels, $\tilde{\boldsymbol{y}}_{\boldsymbol{i}}$ is the calculated output for each pixel, and $\boldsymbol{y}_{\boldsymbol{i}}$ is the original value of each pixel. Furthermore, $\boldsymbol{\delta_\theta}$ and $\boldsymbol{\delta_S}$ provide a measure of the accuracy of the azimuthal angle and size of the metal loss, respectively, whereas the MAE, which is influenced by both, can better represent the overall performance of an NN. A good prediction is, therefore, the one comprising low values for all three metrics.

Fig. 7 depicts the predictions made by the NN algorithm for the simulated cases. The errors calculated based on (12), (13), and (14) are listed in Table VIII the data in the table show that the NN exhibited an acceptable performance in predicting and characterizing the different defects with a maximum calculated error MAE of 0.06.

TABLE VIII
NN PREDICTION ERRORS FOR 18 CASES SIMULATED BY ANSYS MAXWELL

Case Num.	Angle Error (°)	Size Error (%)	MAE
1	−1.03662	17.10526	0.010031
2	−1.60714	−2.72727	0.008488
3	−1.91514	−0.69444	0.010031
4	2.758969	−0.55556	0.01466
5	2.043744	−12.2642	0.020062
6	−2.4679	−10.2041	0.013117
7	2.712514	8.783784	0.01466
8	−3.5332	−4	0.0020062
9	0.371346	−6.08108	0.006944
10	−5.12809	4.72973	0.028549
11	−5.54158	−4.86111	0.060185
12	−4.76064	1.020408	0.030093
13	3.84624	7.986111	0.033179
14	−5.65263	2.777778	0.021605
15	0.893226	0	0.001543
16	2.021365	1.886792	0.003858
17	0.612963	−11.2676	0.006173
18	−0.42965	3.883495	0.003086

For the first five cases, the predictions show either one- or three-pixel rows of wrong estimations, which are equivalent to ∼1 to 3 mm. For cases 6–10, based on the error pixels, we can observe that the predicted holes are accurately located with a slight shift. The largest shift of ∼5° shift occurred in case 10. The highest angle error of ∼5° shifts, when two 2.5-cm holes occur on the same pipe body layer (cases 11–14), was observed in the 11th and 14th cases. Furthermore, the highest estimation error for the hole size, a ∼2 mm^2 drift, is observed in case 13. Finally, the predictions from the last four cases have low angle errors of 1 mm^2 and maximum angle errors of 2°. The percentage

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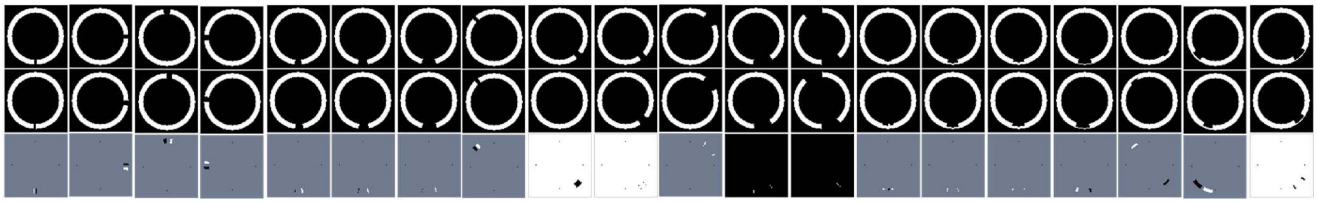


Fig. 10. Performance of the dense NN on hole size and location estimation for experimental (a) case 1, (b) case 2, (c) case 3, (d) case 4, (e) case 5, (f) case 6, (g) case 7, (h) case 8, (i) case 9, (j) case 10, (k) case 11, (l) case 12, (m) case 13, (n) case 14, (o) case 15, (p) case 16, (q) case 17, (r) case 18, (s) case 19, and (t) case 20. The first row shows the ground truth; the second row shows the NN prediction; and the last row represents the error pixels.

size error in case 17 appears severe as the original metal loss is shallow, and a small deviation has resulted in a relatively large percentage of error.

B. Experimental Verification

Fig. 8 portrays a laboratory prototype, which is developed based on the system shown in Fig. 1 and the parameters tabulated in Table VI to validate the ability of the NN to detect different defects experimentally. DC power is supplied by N5771A from KEYSIGHT to the half-bridge inverter that is built using low-$R_{DS(on)}$ silicon carbide MOSFETs C2M0080120D from CREE.

All resonant inductors are developed using enameled wires of 17 AWG for the Tx coil and 31 AWG for the 4 sensor coils. High-power, high-frequency banks of polypropylene film capacitors connected in series with toggle switches are used to separately tune the resonant capacitor for each coil. The outputs of the four sensor resonant circuits are fed to the MSOX6004A oscilloscope from KEYSIGHT. Waveforms are then collected from the oscilloscope by a computer via a LAN cable for processing by the NN algorithm. All experiments are performed in a lab environment without any special pressure, temperature, or EM shielding. Table IX summarizes the 20 cases that have been tested. Fig. 9 shows an example of the sensor coil outputs as a response to a full loss and inner loss. The difference in values between the full and inner loss cases arises because of the elimination of a part of the pipe (about 25% of the pipe body) to machine the inner loss defects.

Fig. 10 illustrates the NN predictions for 20 experimental cases. Similar to the simulated cases, the errors in size and angle predictions, in addition to the MAE, are calculated for all experimental cases using (12), (13), and (14). The calculated errors for the experimental cases are listed in Table X. Based on the conducted experimental work, the minimum full loss defect that could be detected, whether it comes in front of the sensor coils or between two coils, must have at least 10-mm axial length and 4-mm circumferential length. The losses with fewer dimensions are either not detected or detected with high error values, especially when the loss comes between the sensors. However, inner loss defects are detected at any location if the defect had a diameter of 18 mm or more. The inner defects with a diameter of <18 mm are detected with high accuracy only when these defects are in front of one of the sensor coils; they are missed or were detected with a high error rate if they are

TABLE IX
EXPERIMENTAL CASES

Case Num.	Num. of Defects	Defect Type	Defect Dimensions (mm)	Defect Depth (mm)	Defect Location(s)
1	1	Full Loss	4×10	6.02	0°
2	1	Full Loss	6.25×6.25	6.02	90°
3	1	Full Loss	10×4	6.02	180°
4	1	Full Loss	10×10	6.02	270°
5	1	Full Loss	12.5×12.5	6.02	0°
6	1	Full Loss	18.75 x 18.75	6.02	0°
7	1	Full Loss	25×25	6.02	0°
8	1	Full Loss	4×10	6.02	225°
9	1	Full Loss	8.13×8.13	6.02	45°
10	1	Full Loss	12.5×12.5	6.02	45°
11	1	Full Loss	25×25	6.02	135°
12	1	Full Loss	25×25	6.02	15°
13	2	Full Loss	25×25	6.02	20° and 200°
14	1	Inner Loss	10 (Dia.)	2.35	0°
15	1	Inner Loss	13 (Dia.)	4	0°
16	1	Inner Loss	15 (Dia.)	3.4	0°
17	1	Inner Loss	18 (Dia.)	5.5	0°
18	1	Inner Loss	10 (Dia.)	2.35	45°
19	1	Inner Loss	13 (Dia.)	4	315°
20	1	Inner Loss	18 (Dia.)	5.5	45°

TABLE X
NN PREDICTION ERRORS FOR 20 EXPERIMENTAL CASES

Case Num.	Angle Error (°)	Size Error (%)	MAE
1	6.1982	–7.1428	0.0370
2	–4.7272	–1.5811	0.0278
3	–5.4608	4.4225	0.0316
4	–4.8753	6.0285	0.0285
5	2.5884	–2.4117	0.0139
6	1.7529	0.4081	0.0116
7	1.7529	0.4081	0.0116
8	–4.2315	4.1642	0.0216
9	–	–100.0000	0.0540
10	–0.0012	3.2151	0.0062
11	–0.53373	–2.66667	0.003086
12	0.146334	–4.7619	0.005401
13	–1.1966	–1.35135	0.003086
14	–0.5935	0.0000	0.0123
15	–0.4380	8.7719	0.0085
16	–0.6222	6.5217	0.0069
17	–4.4337	0.0000	0.0216
18	173.1436	–27.7777	0.0378
19	28.9825	–23.9130	0.0856
20	0.3515	–32.8358	0.0170

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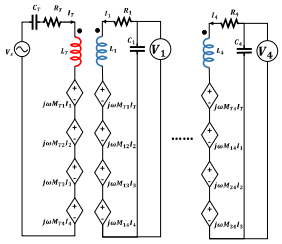


Fig. 11. Circuit simplification by the sinusoidal approximation and the equivalent circuit for the magnetically coupled coils used for analysis.

located between two sensor coils. Accordingly, when a higher resolution is required, the number of sensor coils should be increased.

Experimental testing demonstrated that readings are susceptible to changes in the lift-off distance (i.e., the distance between the coil and pipe wall). This is a well-known concern in the eddy current testing industry and may be reduced by ensuring that all detecting coils interrupt the pipe wall with a uniform amount of space between them.

VI. Conclusion

In this study, EM resonance coupling and ML techniques were used in combination to develop an innovative tool to inspect pipelines and wellbore tubing for metal loss. The proposed tool featured a simple structure and inexpensive components, where the transmitting side consisted of a single *LC* series resonant circuit, whereas the sensor side comprised four parallel resonant circuits. The outputs of the sensor coils were fed to an ANN to predict different types of defects. The ANN was trained using the results generated from the FDTD EM solver, which simulated sensor readings for different scenarios of defects. The ability of the proposed tool to detect and completely characterize partial and full metal loss with different sizes and locations of defects was confirmed using simulation and experimental tests. Full losses with an axial length of 10 mm and a circumferential length of 4 mm or higher and inner losses with a diameter of 18 mm or more were detected with minimal error rates.

Future work will focus on detecting outer metal loss and the possibility of inspecting multiple nested pipes.

Appendix A

This appendix describes the driving of the output voltage from the sensor coils (2). The circuit configuration shown in Fig. 1(d) is rearranged to match the equivalent circuit shown in Fig. 11. The inverter shown in Fig. 1(d) produces a square-wave voltage. This voltage is applied to the input terminal of the *LC* series resonant circuit. The resonant circuit frequency is tuned to the fundamental component of the inverter square wave, i.e., to the switching frequency (*f*). Consequently, the resonant circuit current, as well as the resonant circuit output voltage, essentially has sinusoidal waveforms at the fundamental frequency *f*, with negligible harmonics [28]. Accordingly, the sinusoidal approximation is first considered, where input voltage v_s was obtained as follows:

$$v_s = \frac{2V_D}{\pi}\ \sin(\omega t) = V_s\ \sin(\omega t)\,. \tag{15}$$

The coupling effect of sensor coils on the Tx side can be represented by dependent voltage sources with values $j\omega M_{Ti} I_i$. Similarly, the coupling between one of the sensor coils, the transmitter, and the other sensor coils can be represented as dependent voltage sources with values $j\omega M_{Ti} I_T$ and $j\omega M_{ij} I_j$, respectively. From Kirchhoff's voltage law, we have the following:

$$V_s = Z_T\ I_T + j\omega M_{T1} I_1 + j\omega M_{T2} I_2 + j\omega M_{T3} I_3 + j\omega M_{T4} I_4 \tag{16}$$

$$0\ = Z_1\ I_1 + j\omega M_{T1} I_T + j\omega M_{12} I_2 + j\omega M_{13} I_3 + j\omega M_{14} I_4 \tag{17}$$

$$0\ = Z_2\ I_2 + j\omega M_{T2} I_T + j\omega M_{12} I_1 + j\omega M_{23} I_3 + j\omega M_{24} I_4 \tag{18}$$

$$0\ = Z_3\ I_3 + j\omega M_{T3} I_T + j\omega M_{13} I_1 + j\omega M_{23} I_2 + j\omega M_{34} I_4 \tag{19}$$

$$0\ = Z_4\ I_4 + j\omega M_{T4} I_T + j\omega M_{14} I_1 + j\omega M_{24} I_2 + j\omega M_{34} I_3 \tag{20}$$

where $Z_k = R_k\ + j\omega L_k + {}^{1}\!/_{j\omega C_k}, k\ =\ T,\ 1,\ 2,\ 3,\ 4.$

The output voltages $V_1, V_2,\ V_3,$ and V_4 are given as follows:

$$V_i = \frac{-I_i}{j\omega C_i}. \tag{21}$$

By substituting (17)–(20) into (21), we obtain the following:

$$V_1 = \frac{j\omega\left(M_{T1} I_T + M_{12} I_2 + M_{13} I_3 + M_{14} I_4\right)}{j\omega C_1 R_1 - \omega^2 C_1 L_1 + 1} \tag{22}$$

$$V_2 = \frac{j\omega\left(M_{T2} I_T + M_{12} I_1 + M_{23} I_3 + M_{24} I_4\right)}{j\omega C_2 R_2 - \omega^2 C_2 L_2 + 1} \tag{23}$$

$$V_3 = \frac{j\omega\left(M_{T1} I_T + M_{13} I_1 + M_{23} I_2 + M_{34} I_4\right)}{j\omega C_3 R_3 - \omega^2 C_3 L_3 + 1} \tag{24}$$

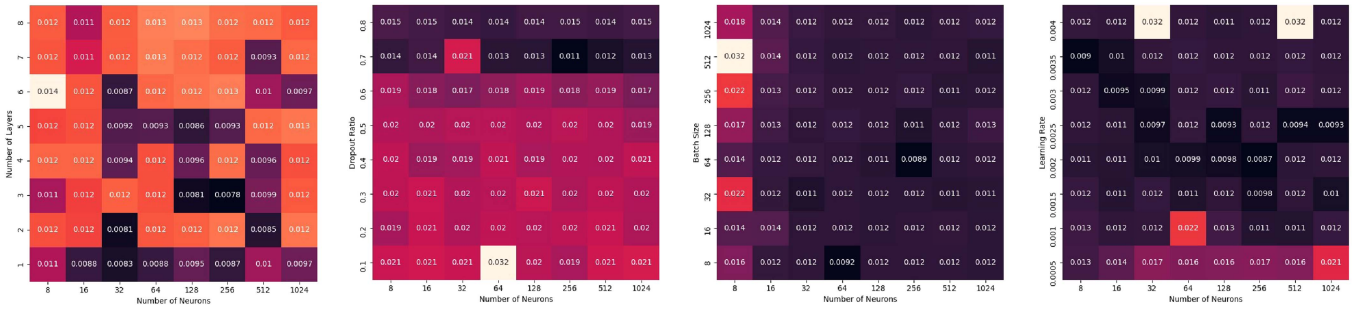
$$V_4 = \frac{j\omega\left(M_{T4} I_T + M_{14} I_1 + M_{24} I_2 + M_{34} I_3\right)}{j\omega C_4 R_4 - \omega^2 C_4 L_4 + 1}. \tag{25}$$

The above-mentioned relations can be represented as shown in (2). The inductance and mutual inductance values can be acquired using ANSYS Maxwell simulations and inputted into a set of equations that can be solved as a linear equation system using MATLAB.

Appendix B

This appendix displays 2-D heatmaps that visualize the performance of each set of hyperparameters in the grid search on the test dataset using ANNs. Fig. 12 shows the performance by plotting the variation of several hyperparameters against the number of neurons in each layer, such as the number of dense layers, dropout ratio, batch size, and learning rate, in terms of average MAEs. The ranges of the tested hyperparameters are shown in Table III, and the set resulting in the

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(a) (b) (c) (d)

Fig. 12. Heatmaps of the average ANN MAE on test data by varying following hyperparameters against number of neurons in each layer: (a) number of dense layers of the NN, (b) dropout ratio, (c) batch size, and (d) learning rate.

TABLE XI
EFFECTS OF OPTIMIZERS ON ANN MAES

	Adam	Adagrad	SGD	RMSProp
MAE	0.0393	0.0466	0.1661	0.1661

lowest overall MAE is selected for the ANN. In addition, the effects of using different optimizers on the ANN are listed in Table XI.

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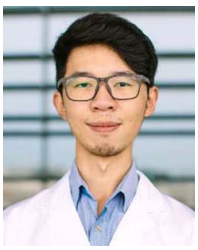
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Tarek M. Mostafa received the B.S. degree in electronics and electrical communication engineering from the Arab Academy for Science, Technology and Maritime Transport (AAST), Giza Governorate, Egypt, in 2011, and the M.S. degree in electronics and materials engineering and the Ph.D. degree in electronics engineering both from Kyushu University, Fukuoka, Japan, in 2015 and 2018, respectively.

He is currently a Research Specialist with the King Abdullah University of Science and Technology, Thuwal, Saudi Arabia. His research interests include near-field wireless power transfer, nondestructive testing, and e-waste recycling and its economics.

Dr. Mostafa was the recipient of the Interdisciplinary Graduate School of Engineering Sciences Dean's Award, the Green Asia Fellowship, and an AAST full scholarship for postgraduate studies.



Guang Ooi received the B.Sc. degree in aeronautics and astronautics engineering from National Cheng Kung University, Tainan, Taiwan, in 2017, and the M.Sc. degree in electrical and computer engineering in 2019 from the King Abdullah University of Science and Technology, Thuwal, Saudi Arabia, where he is currently working toward the Ph.D. degree in electrical and computer engineering under the guidance of Professor Shehab.

His research interests include neural networks, electromagnetic inversion, and drilling simulation.



M. Burak Ozakin (Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees in electronics engineering from Gebze Technical University, Gebze, Turkey, in 2009, 2011, and 2017, respectively.

He is currently a Postdoctoral Fellow with the King Abdullah University of Science and Technology, Thuwal, Saudi Arabia. His research interests include numerical and analytical solutions to electromagnetic problems in the time and frequency domains.



Moutazbellah Khater received the B.Sc. degree in electronics and communication engineering from Al-Ahliyya Amman University, Amman, Jordan, in 2005, the M.Sc. degree (with distinction) in mobile communications from Loughborough University, Loughborough, U.K., in 2007, and the Ph.D. degree in electrical engineering and electronics from the University of Liverpool, Liverpool, U.K., in 2014.

In 2021, he joined the King Abdullah University of Science and Technology, Thuwal, Saudi Arabia, as a Research Scientist . He has more than 14 years of industrial experience in electronics design and manufacturing, power electronics, and automotive electronics.



Mohamed Larbi Zeghlache received the M.Sc. degree in reservoir engineering and field development from the French Petroleum Institute, Paris, France, in 2002.

He is a Senior Researcher in the Production Technology Team of the Saudi Aramco Advanced Research Centre, Dhahran, Saudi Arabia. Since joining Saudi Aramco, he has served as a Senior Petrophysicist, providing expertise in formation evaluation and well placement in Ghawar carbonates and Central Arabia clastics, with 21 years of experience in the oil and gas industry, he has worked with both Schlumberger and Halliburton.



Hakan Bagci (Senior Member, IEEE) received the B.S. degree in electrical and electronics engineering from Bilkent University, Ankara, Turkey, in 2001, and the M.S. and Ph.D. degrees in electrical and computer engineering from the University of Illinois at Urbana-Champaign, Urbana, IL, USA, in 2003 and 2007, respectively.

Since August 2009, he has been with the King Abdullah University of Science and Technology, Thuwal, Saudi Arabia, where he is currently a Professor of Electrical Engineering. His research interests include various aspects of theoretical and applied computational electromagnetics, with an emphasis on well-conditioned frequency- and time-domain integral equation formulations.



Shehab Ahmed (Senior Member, IEEE) received the B.Sc. degree in electrical engineering from Alexandria University, Alexandria, Egypt, in 1999, and the M.Sc. and Ph.D. degrees in electrical engineering from the Department of Electrical and Computer Engineering, Texas A&M University, College Station, TX, USA, in 2000 and 2007, respectively.

From 2001 to 2007, he was with Schlumberger Technology Corporation, Houston, TX, USA, where he developed downhole mechatronic systems for oilfield service products. From 2007 to 2018, he was with Texas A&M University, Qatar. In 2018, he joined the King Abdullah University of Science and Technology, Thuwal, Saudi Arabia, where he is currently a Professor of Electrical Engineering with the CEMSE Division. His research interests include mechatronics, solid-state power conversion, and electric machines.

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